Example 5.22

Find the foci, vertices and length of major and minor axis of the conic $4x^2+36y^2+40x-288y+532=0$.

Solution

Completing the square on x and y of $4x^2+36y^2+40x-288y+532=0$,

$$4(x^2+10x+25-25)+36(y^2-8y+16-16)+532 = 0, \text{ gives}$$

$$4(x^2+10x+25)+36(y^2-8y+16) = -532+100+576$$

$$4(x+5)^2+36(y-4)^2 = 144.$$

Dividing both sides by 144, the equation reduces to

$$\frac{(x+5)^2}{36}+\frac{(y-4)^2}{4}=1.$$

This is an ellipse with centre $(-5,4)$, major axis is parallel to x-axis, length of major axis is 12 and length of minor axis is 4. Vertices are $(1,4)$ and $(-11,4)$.

$$\text{Now, } c^2 = a^2-b^2 = 36-4 = 32$$

$$\text{and } c = \pm 4\sqrt{2}.$$

Then the foci are $\left(-5-4\sqrt{2},4\right)$ and $\left(-5+4\sqrt{2},4\right)$.

Length of the major axis $= 2a = 12$ units and
the length of the minor axis $= 2b = 4$ units.

Example 5.23

For the ellipse $4x^2+y^2+24x-2y+21=0$, find the centre, vertices, and the foci. Also prove that the length of latus rectum is 2.

Solution

Rearranging the terms, the equation of ellipse is

$$4x^2+24x+y^2-2y+21 = 0$$

$$\text{That is, } 4\left(x^2+6x+9-9\right)+\left(y^2-2y+1-1\right)+21 = 0,$$

$$4(x+3)^2-36+(y-1)^2-1+21 = 0,$$

$$4(x+3)^2+(y-1)^2 = 16,$$

$$\frac{(x+3)^2}{4}+\frac{(y-1)^2}{16} = 1.$$

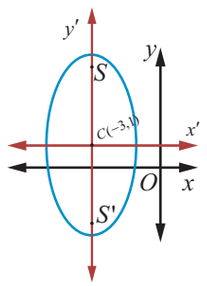


Fig.5.37

Centre is $(-3,1)$ $a=4$, $b=2$, and the major axis is parallel to y-axis

$$c^2 = 16-4 = 12$$

$$c = \pm 2\sqrt{3}.$$

Therefore, the foci are $\left(-3,2\sqrt{3}+1\right)$ and $\left(-3,-2\sqrt{3}+1\right)$.

Vertices are $\left(3,\pm 4+1\right)$. That is the vertices are $(3,5)$ and $(3,-3)$, and

the length of Latus rectum $= \frac{2b^2}{a} = 2$ units. (see Fig. 5.37)

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